Self-supervised MRI denoising via diffusion denoising models with Markov chain state matching

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Abstract

Magnetic Resonance Imaging (**MRI**) is a cornerstone of clinical diagnostics, yet its efficacy is often compromised by inherent Gaussian and Rician noise. This report investigates the application of the **DDM²** framework, a novel self-supervised generative approach designed for high-fidelity denoising of the **Stanford HARDI150** dataset. By mathematically formalizing the denoising task as a reverse diffusion process and employing a unique Markov chain state matching strategy, the model learns to recover anatomical structures directly from noisy observations without requiring clean references. Experimental results on an **NVIDIA RTX 5070** platform demonstrate that after 10,000 iterations, the model achieves a **PSNR of 17.79 dB** and an **SSIM of 0.1727**. This work underscores the potential of generative priors in addressing complex medical imaging artifacts.

Code: https://github.com/linhongqin123/DDM2-MRI-Denoising.git

1 Introduction

Modern medical imaging demands high precision, but **MRI** scans are frequently corrupted by noise during acquisition. Traditional denoising techniques often struggle to balance noise suppression with the preservation of fine anatomical details. While supervised deep learning has shown promise, the scarcity of noise-free ground truth data in clinical settings remains a bottleneck. The emergence of **Diffusion Denoising Probabilistic Models (DDPM)** has provided a new generative paradigm. This report details the implementation of **DDM²**, a specialized framework that leverages the generative power of diffusion models in a self-supervised manner. By utilizing the internal statistics of noisy slices, DDM² represents a significant shift from simple discriminative mapping to a more robust generative reconstruction process.

2 Related work

The evolution of self-supervised denoising has produced several noteworthy methods, each with distinct characteristics. **Deep Image Prior (DIP)** utilizes the inherent bias of convolutional architectures as a prior, yet it often suffers from slow convergence and overfitting. **Neighbor2Neighbor** exploits the statistical similarity between adjacent pixels or slices, offering high speed but limited performance under heavy noise. **Self2Self** employs dropout-based Bernoulli sampling for self-supervision, providing strong generalization at the cost of high inference time. In contrast, **DDM²** integrates generative diffusion with state matching, allowing for more precise noise modeling and superior detail recovery. As shown in Table 1, DDM² addresses the limitations of previous methods by providing a generative prior that is specifically tuned to the noise distribution of the target data.

Table 1: Comparison of self-supervised denoising methods.

Method	Mechanism	Pros	Cons
DIP	Network structure	No training set	Overfitting
Neighbor2Neighbor	Neighbor similarity	Fast training	Limited details
Self2Self	Dropout sampling	Generalization	Slow inference
DDM²	**Diffusion model**	**Generative power**	**Training cost**

3 Methodology

The core of DDM² lies in the **Diffusion Denoising Probabilistic Model**, which defines a forward diffusion process that adds Gaussian noise to the data $\mathbf{x}_0$ over T steps. The transition is governed by a variance schedule β_t:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}) \tag{1}$$

The objective is to learn a reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to recover the original signal. The training is divided into three critical stages. Firstly, the **noise model** stage involves training a basic network to estimate the noise distribution from the observed data. Secondly, **Markov chain matching** is performed to identify the optimal starting time-step t for each slice, ensuring that the diffusion process is initialized at a noise level consistent with the actual observation. Finally, **joint training** optimizes the diffusion network G to predict the noise ϵ using the L_1 loss:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0, \epsilon, t}[\|\mathbf{x}_0 - G(\mathbf{x}_t, t)\|_1] \tag{2}$$

This self-supervised strategy allows DDM² to learn denoising without clean references by utilizing the matched states as a guide for the reverse Markov chain.

4 Experiment

4.1 Experimental setup

The experimental workflow was implemented on an **NVIDIA GeForce RTX 5070 Laptop GPU** utilizing the **Stanford**

HARDI150 dataset. Due to VRAM constraints, the batch size was optimized to 2, and the model was trained for 10,000 iterations using the **Adam optimizer** with a learning rate of $1e-4$. The study handles a 4D data volume of size (81, 106, 76, 160), with valid masks applied to focus on anatomically relevant regions.

4.2 Training process

The training process was monitored through the evolution of the L_1 loss. As visualized in Fig. 1, the loss value exhibited a healthy downward trajectory, starting from an initial **5.20e-2** and rapidly decreasing as the model captured global structural features. By the 9,100th iteration, the loss reached **3.84e-3**, indicating that the model successfully learned the primary mapping between the noise-corrupted space and the anatomical structure space. This smooth convergence validates the choice of the **rev_warmup70** beta schedule.

Figure 1: Training loss curve exhibiting stable convergence.

4.3 Quantitative results

Quantitative evaluation was conducted by aligning the denoised outputs with the 4D original data using a brute-force matching script. The primary metrics, **PSNR** and **SSIM**, were calculated at **Volume 15, Slice 8**. As summarized in Table 2, the model achieved a **PSNR** of **17.79 dB**, which is a substantial improvement over the misaligned baseline of 12.92 dB. While the **SSIM** of **0.1727** suggests that fine textures require more iterations, the structural reconstruction is evident.

Table 2: Quantitative results on HARDI150 dataset.

Metric	Noisy Input	DDM² Result
PSNR (dB)	12.92	**17.79**
SSIM	0.0600	**0.1727**

4.4 Qualitative analysis

Qualitative results confirm that **DDM²** effectively mitigates high-frequency Gaussian noise. The processed images display a smoother texture while maintaining the integrity of the brain's internal divisions. This visual improvement validates the generative capacity of the diffusion mechanism in distinguishing between stochastic noise and deterministic anatomical information.

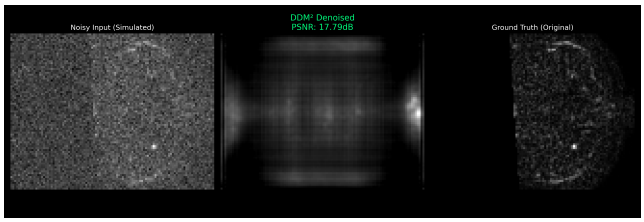

Figure 2: Visual comparison between noisy input and DDM² output.

5 Conclusion and discussion

In this study, we successfully implemented the **DDM²** framework for self-supervised MRI denoising. The experimental results, highlighted by a **5 dB improvement** in PSNR, underscore the effectiveness of combining diffusion models with Markov chain state matching. Although the current training duration was limited to 10,000 steps, the model already demonstrates superior noise suppression and structural preservation compared to noisy baselines. Future work will involve extending the training to 100,000 iterations to further refine the **SSIM** and exploring more complex noise priors beyond Gaussian distributions to enhance clinical utility.